

GESTALTEN

DAY 1

Overview talk (Aoki, 60 min).

Lecture 1: Basics of presentable $(\infty, 1)$ -categories (60 min). Reference: [Sch26, Lecture II]

In general, ∞ -categories that we would like to consider (e.g., the derived $(\infty, 1)$ -category $D(\mathbb{Z})$ of $\mathbb{Z}$ -modules) are not essentially small, and there is a size issue. However, the $(\infty, 1)$ -category $D(\mathbb{Z})$ is controlled by the small subcategory $\text{Perf}(\mathbb{Z})$ of perfect complexes. Such ∞ -categories are called *presentable $(\infty, 1)$ -categories*, and they play an important role in the theory of gestalten. More precisely, for each regular cardinal κ , we define the notion of *κ -presentable $(\infty, 1)$ -categories*. The key property of κ -presentable $(\infty, 1)$ -categories is that the $(\infty, 1)$ -category Pr_κ^L of κ -presentable $(\infty, 1)$ -categories is κ -presentable. In other words, we have $\text{Pr}_\kappa^L \in \text{Pr}_\kappa^L$.

Outline:

- (1) Define κ -presentable $(\infty, 1)$ -categories and related notions (e.g., κ -compact objects, $\text{Ind}_\kappa(-)$). See also [Lur09, 5.3, 5.4] or [Kerodon, 04KE].
- (2) Explain the adjoint functor theorem ([Sch26, Theorem 3.1 and Corollary 3.2]).
- (3) Sketch of the proof of $\text{Pr}_\kappa^L \in \text{Pr}_\kappa^L$ ([Sch26, Theorem 2.9]). See also [Aok25, 2.1].

Lecture 2: Algebras in Pr_κ^L (90 min). Reference: [Sch26, Lecture III].

In [Lur09, Lur17], Lurie gives a very convenient framework to do “algebra” in symmetric monoidal $(\infty, 1)$ -categories, such as $D(\mathbb{Z})$. On the other hand, Pr_κ^L admits a natural symmetric monoidal structure (Lurie’s tensor product). Therefore, we can study symmetric monoidal κ -presentable $(\infty, 1)$ -categories using “algebra”.

As explained in Lecture 1, we have $\text{Pr}_\kappa^L \in \text{Pr}_\kappa^L$, and we can regard Pr_κ^L as a commutative algebra object in Pr_κ^L . Thus, we can define the $(\infty, 1)$ -category $2\text{Pr}_\kappa^L := \text{Mod}_{\text{Pr}_\kappa^L}(\text{Pr}_\kappa^L)$. For $C \in 2\text{Pr}_\kappa^L$, we have the Pr_κ^L -module structure $\text{Pr}_\kappa^L \otimes C \rightarrow C$, and as a right adjoint, we get $\text{Hom}_C(x, y) \in \text{Pr}_\kappa^L$ for $x, y \in C$. In other words, we can regard C as a category enriched in $(\kappa$ -presentable) $(\infty, 1)$ -categories (i.e., κ -presentable “ $(\infty, 2)$ -category”) and 2Pr_κ^L as the $(\infty, 1)$ -category of “ $(\kappa$ -presentable) $(\infty, 2)$ -categories”. Inductively, we define the $(\infty, 1)$ -category $n\text{Pr}_\kappa^L$ of “ $(\kappa$ -presentable) (∞, n) -categories” by $n\text{Pr}_\kappa^L := \text{Mod}_{(n-1)\text{Pr}_\kappa^L}(\text{Pr}_\kappa^L)$. From now on, we take $\kappa = \aleph_1$ ¹ and abbreviate $n\text{Pr} = n\text{Pr}_{\aleph_1}^L$. Roughly speaking, $C_n \in \text{CAlg}(n\text{Pr})$

¹In practice, this choice does not affect concrete mathematical arguments.

is an (∞, n) -category. Moreover, each object $D \in C_n$ defines a $(\aleph_1)$ -presentable $(\infty, n-1)$ -category $\mathrm{Hom}_{(n-1)\mathrm{Pr}}(\mathbf{1}, D) \in (n-1)\mathrm{Pr}$, and using this, we can regard $D \in C_n$ as an $(\infty, n-1)$ -category. We define a Stefanich ring as a sequence $C_1 \in C_2 \in \dots$ of symmetric monoidal $\aleph_1$ -presentable (∞, n) -categories such that $C_n \in C_{n+1}$ is a symmetric monoidal unit of C_{n+1} , and we define the $(\infty, 1)$ -category StRing of Stefanich rings by $\mathrm{StRing} = \lim \mathrm{CAlg}(n\mathrm{Pr})$.

Outline:

- (1) Explain limits and colimits in Pr^L and Pr^R ([Sch26, 3.1, 3.2]).
- (2) Explain Lurie's tensor product and show examples ([Sch26, Definition/Proposition 3.6, Example 3.7, Proposition 3.8, and Example 3.12])
- (3) For a symmetric monoidal κ -presentable $(\infty, 1)$ -category C , explain a fully faithful functor

$$\mathrm{CAlg}(C) \rightarrow \mathrm{CAlg}(\mathrm{Mod}_C(\mathrm{Pr}_\kappa^L)); A \mapsto \mathrm{Mod}_A(C),$$

and its right adjoint functor $D \mapsto \mathrm{End}_D(\mathbf{1})$ ([Sch26, Theorem 3.11]).

- (4) Define the $(\infty, 1)$ -category StRing of Stefanich rings ([Sch26, Definition 3.13]).

Lecture 3: Dualizable categories (60 min). Reference: [Sch26, Appendix to Lecture IV].

As in usual theory of algebra, the notion of dualizable objects in $\mathrm{Mod}_V(1\mathrm{Pr})$ is important in higher category theory (cf. [Ef25]). Ramzi's paper [Ram24] is a good reference for dualizable categories.

Outline:

Review the results in Appendix to Lecture IV in [Sch26], especially [Sch26, Theorem 4.13, Definition 4.20, 4.21, Theorem 4.22].

DAY 2

Lecture 4: Quasi-coherent sheaves of higher categories, I (60 min). Reference: [Sch26, Lecture IV].

In this lecture and the next lecture, as a warm up for Lecture 6, we study Stefanich rings obtained from $\mathrm{CAlg}(D(\mathbb{S}))$. For $A \in \mathrm{CAlg}(D(\mathbb{S}))$, let $D(A)$ denote the category of A -modules. For $n \geq 1$, we define inductively $n\mathrm{Pr}_A = \mathrm{Mod}_{(n-1)\mathrm{Pr}_A}(1\mathrm{Pr})$, where $0\mathrm{Pr}_A = D(A)$. Then $(D(A), 1\mathrm{Pr}_A, 2\mathrm{Pr}_A, \dots)$ is a Stefanich ring, and we get a functor $\mathrm{CAlg}(D(\mathbb{S})) \rightarrow \mathrm{StRing}$ ². For $f: A \rightarrow B$ in $\mathrm{CAlg}(D(\mathbb{S}))$, we write $f_n^*: n\mathrm{Pr}_A \rightarrow n\mathrm{Pr}_B$. The functor $f_0^*: D(A) \rightarrow D(B)$ has a $D(A)$ -linear³ right adjoint functor $f_{0*}: D(B) \rightarrow D(A)$, and if B is a countably presented A -algebra, then the functor $f_{0*}: D(B) \rightarrow D(A)$ defines a morphism in $1\mathrm{Pr}_A$ ⁴. However, in general, the functor $f_0^*: D(A) \rightarrow D(B)$ does not necessarily have a left adjoint functor. A surprising

²Note that $n\mathrm{Pr}_A$ is an object of $\mathrm{CAlg}((n+1)\mathrm{Pr})$. Therefore, we can regard $n\mathrm{Pr}_A$ as an $(\infty, n+1)$ -category, and an object $C \in n\mathrm{Pr}_A$ as an (∞, n) -category.

³ $D(A)$ -linearity of the functor f_{0*} means that f_{0*} satisfies the projection formula.

⁴If B is a countably presented A -algebra, then the functor f_{0*} preserves $\aleph_1$ -compact objects.

point in higher category theory is that the situation is much simplified in the case of $n \geq 1$. More precisely, we have the following theorem:

Theorem 0.1 ([Sch26, Corollary 4.6]). *Let $f: A \rightarrow B$ be a countably presented map in $\mathrm{CAlg}(D(\mathbb{S}))$.*

- (1) *For $n \geq 1$, the functor $f_n^*: n\mathrm{Pr}_A \rightarrow n\mathrm{Pr}_B$ has a left adjoint $f_{n\sharp}: n\mathrm{Pr}_B \rightarrow n\mathrm{Pr}_A$ and a right adjoint $f_{n*}: n\mathrm{Pr}_B \rightarrow n\mathrm{Pr}_A$ in $(n+1)\mathrm{Pr}_A$ ⁵.*
- (2) *There is a canonical identification $f_{n*} \cong f_{n\sharp}$.*
- (3) *For $n \geq 0$, $n\mathrm{Pr}_B \in (n+1)\mathrm{Pr}_A$ is dualizable.*

This phenomenon, of functors having identical left and right adjoints, is called *ambidexterity*. Ambidexterity is a unique phenomenon in higher category theory, and it will play an important role in the following lectures.

Finally, by using Mathew’s descendability argument, we find that the functor $\mathrm{CAlg}(D(\mathbb{S})) \rightarrow \mathrm{StRing}$ is a hypersheaf in the fpqc topology. Therefore, we can extend the above construction to fpqc stacks, and we get a functor⁶

$$\mathrm{Shv}_{\mathrm{fpqc}}(\mathrm{CAlg}(D(\mathbb{S}))) \rightarrow \mathrm{StRing}; X \mapsto (D(X), 1\mathrm{Pr}_X, 2\mathrm{Pr}_X, \dots).$$

Outline:

- (1) Define a functor $\mathrm{CAlg}(D(\mathbb{S})) \rightarrow \mathrm{StRing}$, and construct a right adjoint functor $f_{n*}: n\mathrm{Pr}_B \rightarrow n\mathrm{Pr}_A$ of $f_n^*: n\mathrm{Pr}_A \rightarrow n\mathrm{Pr}_B$ for $f: A \rightarrow B$ in $\mathrm{CAlg}(D(\mathbb{S}))$ ([Sch26, Proposition 4.1]).
- (2) Explain “ambidexterity” and prove Theorem 0.1 ([Sch26, 4.2]).
- (3) Review the notion of Mathew’s descendability ([Mat16]), and sketch the proof of fpqc descent ([Sch26, Theorem 4.8]).

Lecture 5: Quasi-coherent sheaves of higher categories, II (90 min). Reference: [Sch26, Lecture V].

In this lecture, we extend Theorem 0.1 from affine schemes to stacks. The main theorem in this lecture is the following:

Theorem 0.2 ([Sch26, Theorem 5.2]). *Let $f: Y \rightarrow X$ be a morphism (satisfying some countability conditions) in $\mathrm{Shv}_{\mathrm{fpqc}}(\mathrm{CAlg}(D(\mathbb{S})))$.*

- (1) *For $n \geq 1$, the functor $f_n^*: n\mathrm{Pr}_X \rightarrow n\mathrm{Pr}_Y$ has a left adjoint $f_{n\sharp}$ in $(n+1)\mathrm{Pr}_X$.*
- (2) *For $n \geq 2$, then the functor $f_{n\sharp}$ is also a right adjoint of f_n^* in $(n+1)\mathrm{Pr}_X$.*
- (3) *For $n \geq 1$, $n\mathrm{Pr}_Y \in (n+1)\mathrm{Pr}_X$ is dualizable.*

When $n = 0$, the functor $f_0^*: D(X) \rightarrow D(Y)$ usually does not admit a left adjoint $f_{0\sharp}$ in $1\mathrm{Pr}_X$. In terms of 6-functor formalisms, if f is “étale”, then f_0^* admits a left

⁵We regard the (∞, n) -category $n\mathrm{Pr}_B$ as $f_{n+1,*}\mathbf{1} \in (n+1)\mathrm{Pr}_A$, where $f_{n+1,*}: (n+1)\mathrm{Pr}_B \rightarrow (n+1)\mathrm{Pr}_A$ is a right adjoint of the functor $f_{n+1}^*: (n+1)\mathrm{Pr}_A \rightarrow (n+1)\mathrm{Pr}_B$ of presentable $(\infty, 1)$ -categories.

⁶In general, this $D(X)$ might be different from the usual $D(X)$ because of the difference of limits in $1\mathrm{Pr} = \mathrm{Pr}_{\mathbb{N}_1}^L$ and Pr^L . However, if X is a countable colimits of affine schemes, then there is no difference between these two $D(X)$.

adjoint in $1\mathrm{Pr}_X$. Similarly, when $n = 0$, the functor $f_0^*: D(X) \rightarrow D(Y)$ usually does not admit a right adjoint f_* in $1\mathrm{Pr}_X$, and if f is “proper”, then f_0^* admits a right adjoint in $1\mathrm{Pr}_X$. Therefore, roughly speaking, Theorem 0.2 claims that for $n \geq 1$ “all maps are étale”, while for $n \geq 2$ “all maps are proper”.

Outline:

Explain the details of [Sch26, Theorem 5.2, Corollary 5.3, 5.4, 5.5].

Lecture 6: Affine, Proper/Prim, and Étale/Suave (150 min). Reference: [Sch26, Lecture VI].

In this lecture, we study the phenomena appearing in the previous two lectures more systematically for arbitrary Stefanich rings.

Outline:

- (1) Explain the construction in [Sch26, Proposition 6.2].
- (2) Define the notion of n -affine, and prove their basic properties ([Sch26, Proposition 6.6]).
- (3) Define the notion of n -prim and n -proper, and prove their basic properties ([Sch26, Proposition 6.10, Proposition 6.13, Lemma 6.15]). Explain the relationship between 0-properness and rigidity in the sense of Gaitsgory-Rozenblyum ([Sch26, Proposition 6.14]). Assuming [Sch26, Corollary 8.14], sketch the proof of [Sch26, Proposition 8.1].
- (4) Define the notion of n -suave and n -étale, and prove their basic properties ([Sch26, Proposition 6.18, Proposition 6.20]).
- (5) Explain the relationship between *primness* and *suaveness* (ambidexterity, [Sch26, Proposition 6.21, Lemma 6.22]).

DAY 3

Lecture 7: Gestalten (90 min). Reference: [Sch26, Lecture VII].

We define the $(\infty, 1)$ -category Gest of gestalten by $\mathrm{Gest} := \mathrm{StRing}^{\mathrm{op}}$, and we write an object in Gest corresponding to $A \in \mathrm{StRing}$ as $\mathrm{Gest}(A)$. The main theorem in this lecture is the following:

Theorem 0.3 ([Sch26, Theorem 7.1]). *Let A be a Stefanich ring, and $n \geq 0$. Let $\mathrm{StRing}_{A'}^{n\text{-ét}}$ be the $(\infty, 1)$ -category of n -étale A -algebras. Then $(\mathrm{StRing}_{A'}^{n\text{-ét}})^{\mathrm{op}}$ is a countably $(\infty, 1)$ -topos. In other words, $\mathrm{Ind}_{\aleph_1}(\mathrm{StRing}_{A'}^{n\text{-ét}})^{\mathrm{op}}$ is an $(\infty, 1)$ -topos⁷.*

Outline:

- (1) Review (countably) $(\infty, 1)$ -topoi, especially Giraud’s axiom ([Sch26, Theorem 7.2, Proposition 7.3]).
- (2) Prove [Sch26, Proposition 7.5].
- (3) Prove Theorem 0.3 ([Sch26, Theorem 7.1, Theorem 7.7]).
- (4) Explain the descent of affineness (resp. primness, properness, suaveness, étaleness) ([Sch26, Proposition 7.9]).

⁷In practice, we can work with a sufficiently large fixed n (e.g., $n = 10$).

Lecture 8: Descendability in higher categories (90 min). Reference: [Sch26, Appendix to Lecture VIII].

In this lecture, we give a criterion for whether a map $f: X \rightarrow Y$ in $\mathcal{G}\text{est}$ is surjective in practical situations. This criterion can be regarded as a generalization of Mathew’s descendability to higher categories.

Outline:

- (1) Prove [Sch26, Proposition 8.6, 8.7, 8.8, 8.10, 8.11, Lemma 8.13, Corollary 8.14].
- (2) Explain the relation with the notion of Mathew’s descendability ([Sch26, Corollary 8.15, 8.16]).

Lecture 9: Examples (90 min). Reference: [Sch26, Lecture VIII, IX].

In this lecture, we give examples of $\mathcal{G}\text{estalten}$ (e.g., $\mathbb{A}^1$, $\mathbb{G}_a$, GL_1 , $\mathbb{G}_m$, $\mathbb{P}^1$, $*/\mathbb{G}_m$, X_{dR}). It is also possible to construct $\mathcal{g}\text{estalten}$ from a general 6-functor formalism.

Outline:

- (1) Briefly introduce examples in [Sch26, Lecture VIII]. Note that [Sch26, Proposition 8.1] was already explained in Lecture 6.
- (2) Explain the construction of $\mathcal{g}\text{estalten}$ from a general (small) 6-functor formalism ([Sch26, Proposition 9.3, 9.4, 9.5]). Briefly explain the extension of the above construction to the stacks ([Sch26, Proposition 9.7, Corollary 9.8]).

DAY 4

Lecture 10: Cartier duality (120 min). Reference: [Sch26, Lectures X, XI, XII].

In this lecture, we study Cartier duality in terms of $\mathcal{g}\text{estalten}$. There are many examples of certain dualities between commutative group objects in the category of “geometric spaces”.

Example 0.4. (1) Let R be a classical ring, and let $\mathbb{Z} = \bigsqcup_{\mathbb{Z}} \text{Spec } R$ and $\mathbb{G}_m =$

$\text{Spec } R[t^{\pm 1}]$. Then there are natural equivalences

$$\begin{aligned} D_{\text{qc}}(\mathbb{Z}) &\cong D_{\text{qc}}(*/\mathbb{G}_m), \\ D_{\text{qc}}(*/\mathbb{Z}) &\cong D_{\text{qc}}(\mathbb{G}_m). \end{aligned}$$

- (2) Cartier: Let R be a noetherian ring, and let G be a finite flat commutative group scheme over $\text{Spec } R$. Then $\mathcal{O}(G)^* = \text{Hom}(\mathcal{O}(G), R)$ has a natural commutative algebra structure and a cocommutative coalgebra structure induced from the group structure of G and the algebra structure of $\mathcal{O}(G)$. We set $G^* = \text{Spec } \mathcal{O}(G)^*$, which is also a finite flat commutative group scheme over $\text{Spec } R$. Then there are natural equivalences

$$\begin{aligned} D_{\text{qc}}(G) &\cong D_{\text{qc}}(* / G^*), \\ D_{\text{qc}}(* / G) &\cong D_{\text{qc}}(G^*). \end{aligned}$$

- (3) Amice: Let $\mathbb{D}$ be the rigid analytic open unit disc over $\mathrm{Spa} \mathbb{Q}_p$, which we regard as an analytic stack over $\mathrm{AnSpec} \mathbb{Q}_{p,\square}$. Let $\mathbb{Z}_p^{\mathrm{la}} = \mathrm{AnSpec} C^{\mathrm{la}}(\mathbb{Z}_p, \mathbb{Q}_p)_{\square}$, where $C^{\mathrm{la}}(\mathbb{Z}_p, \mathbb{Q}_p)$ is the $\mathbb{Q}_{p,\square}$ -algebra of locally analytic functions on $\mathbb{Z}_p$ with values in $\mathbb{Q}_p$. Then there are natural equivalences

$$D(1 + \mathbb{D}) \cong D(*/\mathbb{Z}_p^{\mathrm{la}}),$$

$$D(*/(1 + \mathbb{D})) \cong D(\mathbb{Z}_p^{\mathrm{la}}).$$

The goal of this lecture is to give a unified explanation of these examples as a duality of gestalten.

Outline:

- (1) Explain examples of classical dualities (some of [Sch26, Examples 10.1-7]).
- (2) Explain the statement of [Sch26, Theorem 10.8], and sketch the proof (especially [Sch26, Theorems 11.1, 12.3, and 12.4, Remark 12.5]).
- (3) Explain the interpretation of the above examples in terms of gestalten. (Feel free to add other examples as you like.)

Advanced topics, I (Aoki, 120 min).

DAY 5

Advanced topics, II (Aoki, 180 min).

REFERENCES

- [Aok25] Ko Aoki, *Higher presentable categories and limits*, arXiv preprint arXiv: [2510.13503](#) (2025).
- [Efi25] Alexander I. Efimov, *K-theory and localizing invariants of large categories*, arXiv preprint arXiv: [2405.12169](#) (2025).
- [Lur09] Jacob Lurie, *Higher topos theory*, Annals of Mathematics Studies, vol. 170, Princeton University Press, Princeton, NJ, 2009.
- [Lur17] ———, *Higher algebra*, <https://www.math.ias.edu/~lurie/papers/HA.pdf>, 2017.
- [Mat16] Akhil Mathew, *The Galois group of a stable homotopy theory*, Adv. Math. **291** (2016), 403–541.
- [Ram24] Maxime Ramzi, *Dualizable presentable ∞ -categories*, arXiv preprint arXiv: [2410.21537](#) (2024).
- [Sch26] Peter Scholze, *Geometry and higher category theory*, <https://people.mpim-bonn.mpg.de/scholze/Gestalten.pdf>, 2026.
- [Kerodon] Jacob Lurie, *Kerodon*, <https://kerodon.net>.